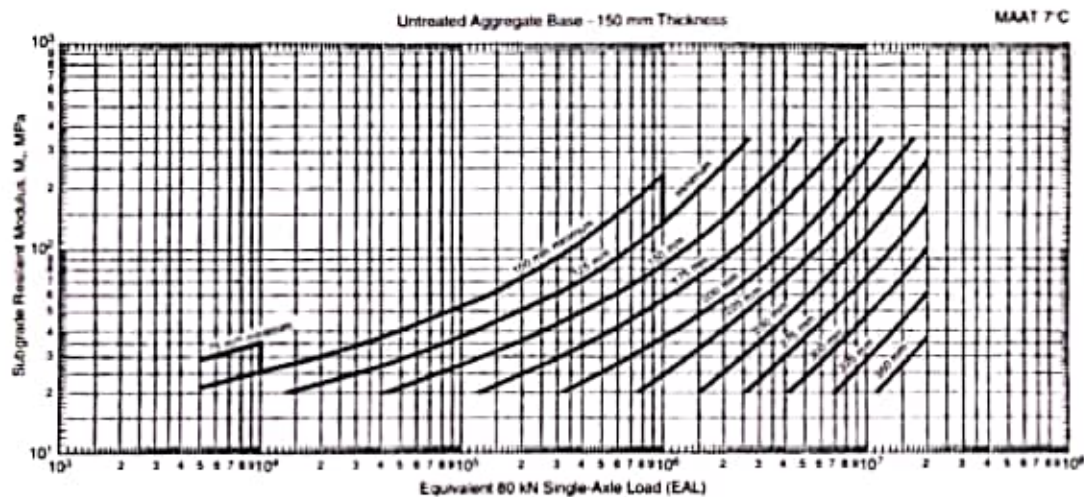


How will the design of an asphalt concrete pavement with an untreated granular base of 300 mm thickness monograph vary if it is required to accommodate 3 million ESALs without failing in fatigue cracking or rutting. Consider the subgrade resilient modulus to be 80 MPa, and the MAAT is 7 °C. AI Thickness Design Nomograph.



- ☐ a. The thickness will be greater than 200 mm.
- ☐ b. The thickness will be 200 mm
- ☐ c. The thickness will be between 130 mm and 200 mm.
- ☐ d. The thickness will be 130 mm.

Pitched roofs are NOT preferred in areas where

- ☐ a. Terrace is required
- ☐ b. There is limited width and simple shape
- ☐ c. There're slopping top surfaces
- ☐ d. Rainfall/snow fall is very heavy

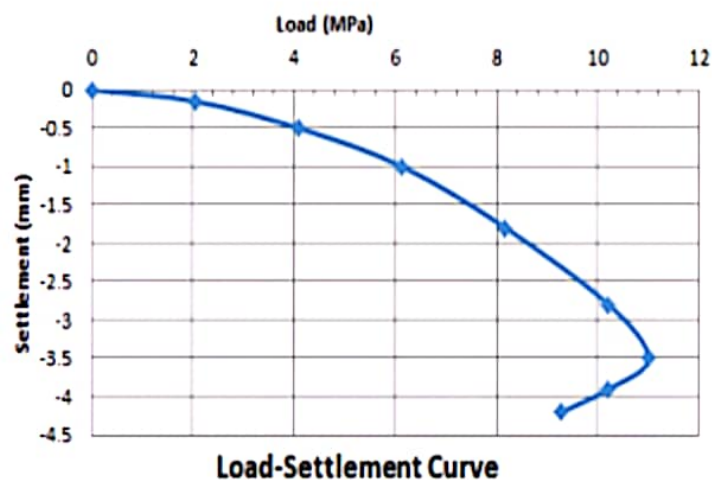
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- ☐ d. The thickness will be 130 mm.

Which one of the following parameters contribute less to the shear strength of reinforced concrete beams

- ☐ a. Longitudinal reinforcement ratio
- ☐ b. Shear reinforcement amount
- ☐ c. Compression reinforcement ratio
- ☐ d. Concrete strength

A 600 mm-diameter and 20 m-long concrete bored pile is constructed using the continuous flight auger technique. The pile passes through 8 m of gravelly sand followed by 8 m of stiff clay, with the bottom 4 m founded in very dense sand. A pile load test was carried out on this pile and the load test data (pressure versus settlement) are summarized in the figure below. Adopting a factor of safety of 2.0, estimate the allowable load on the pile.



- ☐ a. 5.5 MPa
- ☐ b. 11 MPa
- ☐ c. 34 MPa
- ☐ d. 12 MPa

What is NOT the purpose of valuation

- ☐ a. For buying or selling
- ☐ b. Acquisition
- ☐ c. To get over valued loan
- ☐ d. For Security of loans or mortgage

_____ is combinations of ramps and grade separations designed as a system of interconnecting roadways to separate the turning and through movements at the junction of two or more roads.

- ☐ a. Grade separated intersection
- ☐ b. At-grade intersection
- ☐ c. Channelization.
- ☐ d. Interchange.

A project closeout does not constitute

- ☐ a. Project planning
- ☐ b. Contract closeout
- ☐ c. Project Acceptance
- ☐ d. Project completion

While designing a retaining wall, what remedial measure do you consider if the factor of safety against sliding falls down below the requirement?

- ☐ a. Increase the height of the wall.
- ☐ b. Decrease the height of the wall.
- ☐ c. Decrease the base width of the wall.
- ☐ d. Increase the base width of the wall.

Minimum concrete cover is provided in RC members to provide except

- ☐ a. Increased bond strength
- ☐ b. Increased durability such as protecting the reinforcement from corrosion
- ☐ c. Increased fire resistance
- ☐ d. Increased flexural strength

One of the following is used for the computation of the volume of earthwork in a road project.

- ☐ a. Triangular formula.
- ☐ b. Spherical formula.
- ☐ c. Prismoidal formula.
- ☐ d. Simpson's $5/8$ formula.

An effective and efficient procurement method ensures the following rights, except

- ☐ a. The Right Quality
- ☐ b. The Right Quantity
- ☐ c. The Right profit
- ☐ d. The Right Counterpart

The following is NOT considered as the main sources of dampness in buildings during service

- ☐ a. Water used for concrete curing
- ☐ b. Moisture rising up the walls from the ground
- ☐ c. Rain travel from wall tops
- ☐ d. Rain Beating Against External Walls

One of the following is NOT an environmental issue related to soil boring during site investigation.

- ☐ a. Pneumatic caissons are less health hazard than other types of caisson.
- ☐ b. Construction of sheet pile walls can affect aquatic life.
- ☐ c. Deep excavation can cause damage on nearby buildings.
- ☐ d. A collapse of struts in cofferdams can cause life lose.

One of the following tasks is NOT covered in quantity surveying

- ☐ a. Preparation of approximate (preliminary) cost estimate at the very early stage of the project
- ☐ b. Preparation of Specification
- ☐ c. Taking measurements of civil works
- ☒ d. Preparation of construction drawing

[Clear my choice](#)

There is a plan to construct a high-rise building on cast-in-situ pile foundation with about 50 m length. Geotechnical site investigation at the site dictates that ground water is located at shallower depth and the top 10 m layer of the site is sand soil followed by clayey sand. Using these data, which of the following is NOT true?

- ☐ a. As the piles are cast-in-situ piles, the quality of construction of the piles can be reduced.
- ☐ b. Use of casings during the construction of the piles is a must.
- ☐ c. Caution must be made while pouring the concrete as concrete segregation can happen.
- ☐ d. As there will be flow of water to the bore hole, it is a good idea to reduce the water cement ratio during the concrete production.

What is the cause of swelling and shrinkage problem in expansive soils ?

- ☐ a. The black color is the cause for the problem.
- ☐ b. The existence of more illite mineral in the soil.
- ☐ c. The existence of more montmorillonite mineral in the soil.
- ☐ d. The existence of more kaolinite mineral in the soil.

The soil state at which its mean effective stress is zero due to the seeping water is called _____.

- ☐ a. Critical state.
- ☐ b. Static liquefaction.
- ☐ c. Densification.
- ☐ d. Consolidation.

All of the following criteria are considered in the external stability analysis of conventional retaining walls except.

- ☐ a. Sliding
- ☐ b. Overturning
- ☐ c. Bearing capacity
- ☐ d. Nail pullout

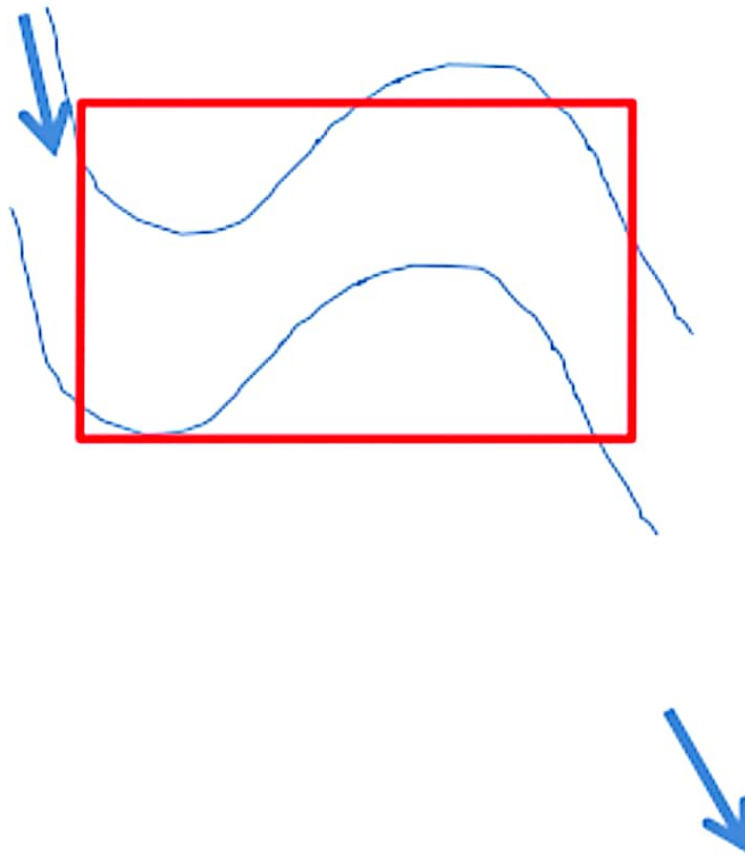
Which one is NOT a common cause failure of dams?

- ☐ a. Structural failure.
- ☐ b. Hydraulic failure.
- ☐ c. Seepage failure.
- ☐ d. Traffic over crest

A square footing is required to carry a dead load of 500 kN and live load of 400 kN column load in a medium-dense sand which has an ultimate bearing capacity of 350 kPa. Adopting a factor of safety of 1.5 for the bearing capacity, analyze and estimate the plan dimensions of the footing.

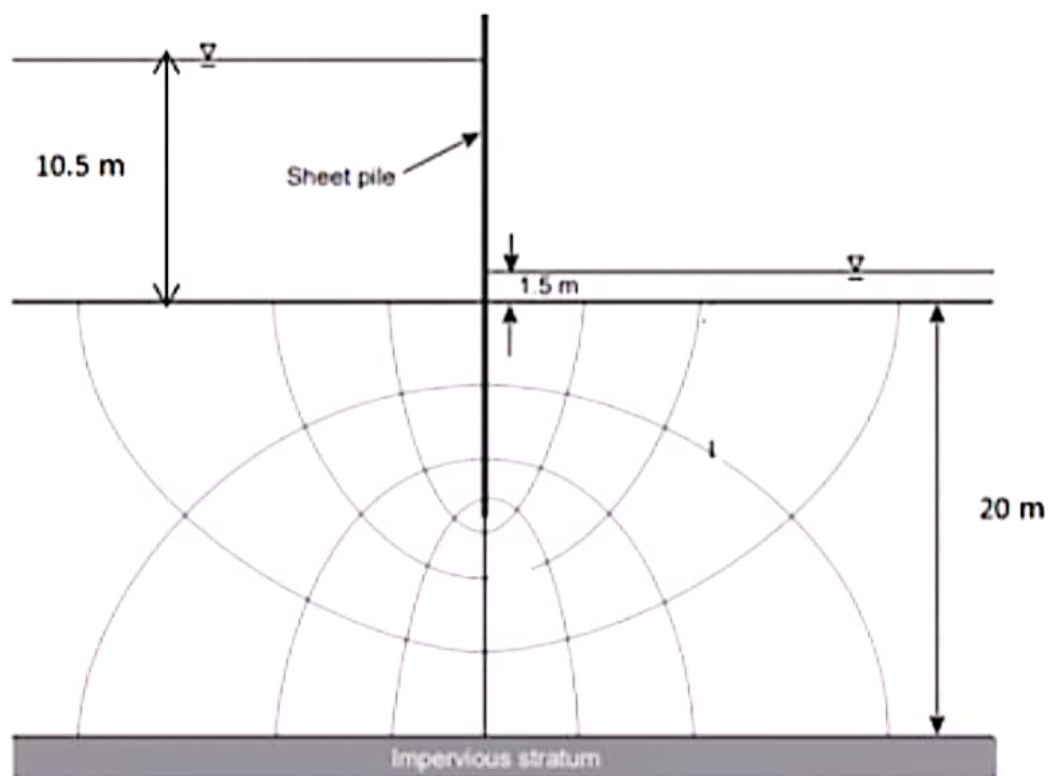
- ☐ a. BxL will be 1.3 x 1.3 m².
- ☐ b. BxL will be 1.6 x 1.6 m².
- ☐ c. BxL will be 1.5 x 1.5 m².
- ☐ d. BxL will be 2 x 2 m².

If the geometry of a potential embankment dam site is on a zigzagging river (S- shape layout) as shown in the figure below, what type of analysis do you recommend if internal tension cracks need to be investigated?



- ☐ a. Two dimensional analysis manually
- ☐ b. Plane stress analysis using software
- ☐ c. Plane strain analysis using software
- ☐ d. Three dimensional analysis manually

A sheet pile wall is driven into sandy silt and seepage takes place through the soil where the coefficient of permeability is 1.5×10^{-4} cm/s. Compute the volume of water per unit width of the wall that flows through the soil in one year using the flow net shown in the figure below.



- ☐ a. 0.437 m³
- ☐ b. 159.65 m³
- ☐ c. 1.820 m³
- ☐ d. 0.0182 m³

One of the following is NOT a negative environmental issue that happens during civil engineering operation.

- ☐ a. Construction of masonry retaining walls intercepts drainage lines.
- ☐ b. Land sliding during excavations.
- ☐ c. Salvage excavated topsoil for landscaping.
- ☐ d. Depletion of construction material.

_____ is a layer of natural soil prepared to receive the stress from sub-base layer.

- ☐ a. Subgrade.
- ☐ b. Asphalt.
- ☐ c. Base course.
- ☐ d. Surface course.

Static wind loading is different from earthquake loading because it depends on

- ☐ a. The geometry/shape of the building
- ☐ b. The stiffness of the structure
- ☐ c. The soil type of the foundation
- ☐ d. The mass of the structure

Which one of the following does NOT contribute to the total settlement in soils?

- ☐ a. Primary consolidation settlement.
- ☐ b. Secondary settlement.
- ☐ c. Immediate settlement.
- ☐ d. Tertiary consolidation settlement.

Sight distance at intersections may be used for the following except.

- ☐ a. Enable approaching vehicle to change the speed.
- ☐ b. Enable parking at road intersections.
- ☐ c. Enable approaching vehicle to stop.
- ☐ d. Enable stopped vehicle to cross a main road.

All of the following describe Level of Service E except.

- ☐ a. Levels of basic elements of comfort and convenience are very poor.
- ☐ b. Traffic flow conditions are best described as unstable.
- ☐ c. The capacity of the highway has been reached.
- ☐ d. Traffic movement is on a 'stop-go' basis.

What is the purpose of drawing flow net?

- ☐ a. To estimate the amount of water flowing.
- ☐ b. To draw flow lines.
- ☐ c. To decrease the amount of water seeping.
- ☒ d. To draw equipotential lines.

[Clear my choice](#)

Which type of formwork material has the highest number of reuse cycles

- ☐ a. Steel
- ☐ b. Plywood
- ☐ c. Timber
- ☐ d. Aluminum

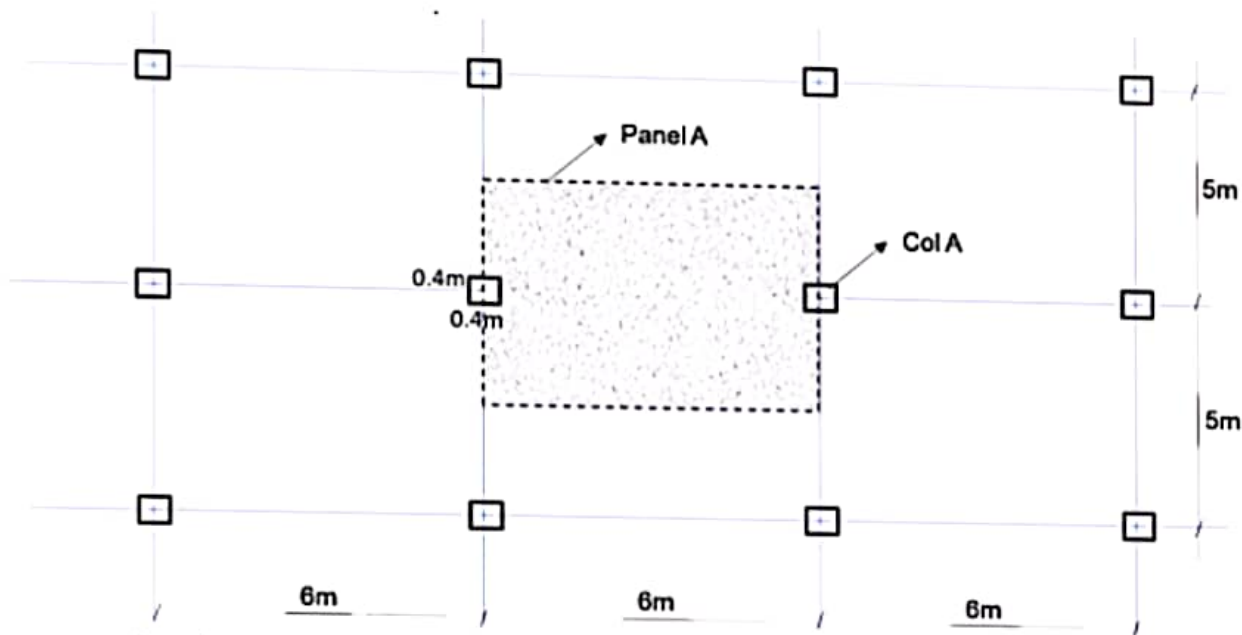
Which of the following is NOT true about the design and analysis of sheet pile retaining walls?

- ☐ a. Sheet pile walls have high salvage values than conventional rigid retaining walls.
- ☐ b. If the depth of excavation is 5 m, one can use cantilever sheet pile wall
- ☐ c. Water pressure due to the water below the dredge line should be ignored during the analysis.
- ☐ d. Anchored sheet pile walls are used in waterfront structures.

Advantages gained by using timber as building structure material is

- ☐ a. High strength to weight ratio
- ☐ b. High fire resistance
- ☐ c. High resistance to warping, bending and cracking
- ☐ d. High resistance to decay

Compute the statical moment in the slab panel A shown in figure below. The flat slab supports a total factored load of 20 kN/m^2 .



- ☐ a. 392 kNm
- ☐ b. 90 kNm
- ☐ c. 375 kNm
- ☐ d. 75 kNm

One of the following design methods is NOT usually used for the structural design of flexible rigid pavement.

- ☐ a. American Association of State Highway and Transportation Officials
- ☐ b. The Portland Cement Association
- ☐ c. The Asphalt Institute
- ☐ d. NCHRP

Which of the following is NOT true about the design and analysis of rigid retaining walls?

- ☐ a. The resultant force on the wall should pass through the first third of the base width of the wall.
- ☐ b. The sum of moments produced by the active earth pressure must be less than the sum of moments produced by the different weights of the wall and backfill soil acting with the wall.
- ☐ c. Usually cohesion-less soil is used as a backfill material in the back of the wall.
- ☐ d. The passive earth pressure below the dredge line provides resistance against sliding.

One of the following is a mechanism used to reduce the negative impact on environment due to construction activities.

- ☐ a. Damping excavated expansive soils during road construction in a very far place.
- ☐ b. Stabilizing expansive soils with granular material.
- ☐ c. Use of cast-in-situ concrete piles over precast concrete piles.
- ☐ d. Paving walkways with white color concrete apron

The environmental factors influencing the strength of concrete at early age of development include, except

- ☐ a. Moisture conditions during curing
- ☐ b. Light conditions during curing
- ☐ c. Pressure conditions during steam curing
- ☐ d. Temperature conditions during curing

The purpose of limiting bridge deflection is to avoid

- ☐ a. Fatigue effects resulting from excessive vibrations
- ☐ b. Excessive deformation stresses in secondary members or connections
- ☐ c. Failure of the bridge
- ☐ d. Undesirable psychological reactions by pedestrians and/or passengers

The advantage gained in using reinforced concrete over steel in the construction of bridges is

- ☐ a. Less affected by corrosion
- ☐ b. Less material usage
- ☐ c. Less prone to cracking
- ☐ d. High resistance to cracking

Which one of the following does NOT contribute to the shear strength of beams without web reinforcement

- ☐ a. Tensile resistance of the longitudinal bar
- ☐ b. Dowel action
- ☐ c. Aggregate interlock
- ☐ d. Shear resistance due the uncracked part of the concrete

What is the purpose of mixing water in concrete

- ☐ a. To increase the workability of fresh concrete
- ☐ b. To Increase the strength of hardened concrete
- ☐ c. To delay the hydration reaction with cement
- ☐ d. To increase the density of hardened concrete

All of the following criteria are considered in the external stability analysis of conventional retaining walls except.

- ☐ a. Sliding
- ☐ b. Bearing capacity
- ☐ c. Overturning
- ☐ d. Nail pullout

What is NOT the primary objective a surveillance programme?

- ☐ a. To assist in the scheduling of routine maintenance.
- ☐ b. To control the amount of water discharging.
- ☒ c. To minimize the possibility of catastrophic failure of the dam.
- ☐ d. To review performance

[Clear my choice](#)

One of the following is NOT a technique used to produce a schedule

- ☐ a. CPM (Critical Path Method)
- ☒ b. PERT (program evaluation and review technique)
- ☐ c. Work break down
- ☐ d. Bar charts

[Clear my choice](#)

Which design and analysis approach do you recommend when you do not have material and site investigation works?

- ☐ a. SEEP/W component of Geoslope.
- ☐ b. Latest software based advanced methods.
- ☐ c. Approximate graphical method.
- ☐ d. SIGMA/W component of Geoslope.

_____ is the bearing capacity of soil that takes into account a required factor of safety for shear and settlement.

- ☐ a. Safe bearing capacity.
- ☐ b. Ultimate bearing capacity.
- ☐ c. Net ultimate bearing capacity.
- ☐ d. Allowable bearing capacity.

For effective (in terms of safety and economy) design of footings, the resultant force from the structure should pass through _____.

- ☐ a. The toe of the footing.
- ☐ b. The heel of the footing.
- ☐ c. The kern.
- ☐ d. The first third.

Which of the following is NOT true about a mass diagram in a road project?

- ☐ a. If the final point on the mass diagram is a positive amount, it indicates a net shortage of earthwork for the project.
- ☐ b. The maximum and minimum points on the mass diagram occur at or near to the grade points on the profile.
- ☐ c. The final point on a mass diagram for a given project gives the overall net amount of earthwork for the entire project.
- ☐ d. Balance line is any horizontal line, cutting off a loop of the mass curve, and is equal to the fill.

The mean wind velocity $V_m(z)$ at a height z above the terrain does not depend on

- ☐ a. Basic wind velocity
- ☐ b. Orography
- ☐ c. Air density
- ☐ d. Terrain roughness

One of the following is NOT a design criterion used in the design of caissons.

- ☐ a. External walls of a concrete caissons is designed to withstand lateral loads.
- ☐ b. Construction joints, if any, in concrete caissons can be placed anywhere along the height of the caisson.
- ☐ c. External walls of a concrete caissons is designed to withstand vertical loads.
- ☐ d. Internal walls are designed to share vertical loads with external walls of the concrete caisson.

Which one of the following construction materials has the highest carbon-foot print (carbon foot-print refers to release of carbon dioxide to the atmosphere)

- ☐ a. Timber
- ☐ b. Earth blocks
- ☐ c. Steel
- ☐ d. Concrete

A 7 m-high smooth, vertical wall retains a horizontal surface granular backfill with unit weight of 18 kN/m^3 and a friction angle of 30° . Find the magnitude and location of the resultant active thrust from the base.

- ☐ a. 147 kN/m , 2.33 m
- ☐ b. 63 kN/m , 2.33 m
- ☐ c. 42 kN/m , 3.00 m
- ☐ d. 126 kN/m , 2.33 m

The seismic force acting on a building structure does NOT depend on

- ☐ a. The mass of the building
- ☐ b. The shape of the building
- ☐ c. The lateral stiffness of the building
- ☐ d. The structural system of the building

_____ is the property of bitumen that permits it to undergo great deformation or elongation.

- ☐ a. Stiffness.
- ☐ b. Viscosity.
- ☐ c. Flexibility.
- ☐ d. Ductility.

A singly reinforced concrete section has dimensions width = 300mm, gross depth = 600mm and effective depth $d=550$ mm. Calculate the neutral axis depth at balanced failure

- ☐ a. 400mm
- ☐ b. 250mm
- ☐ c. 275mm
- ☐ d. 350mm

_____ is the separation or regulation of conflicting traffic movements into definite paths of travel by traffic islands or pavement markings to facilitate the safe and orderly movements of both vehicles and pedestrians.

- ☐ a. Grade separated intersection
- ☐ b. Channelization.
- ☐ c. At-grade intersection
- ☐ d. Interchange.

When two highways or streets cross each other at a different grade, with no connections, the arrangement is referred to as _____.

- ☐ a. Interchange.
- ☐ b. Channelization.
- ☐ c. Grade separated intersection.
- ☐ d. At-grade intersection.

Which one of the following is NOT the purpose of painting

- ☐ a. To protect the surface from weathering effects of the atmosphere
- ☐ b. To prevent cracking
- ☐ c. To protect the decay of wood and corrosion of metals
- ☐ d. To provide a decorative finish

Limit state design is a process that involves, except

- ☐ a. The identification of all potential modes of failure
- ☐ b. Identification of working loads and allowable stresses
- ☐ c. The determination of acceptable levels of safety against occurrence of each limit state
- ☐ d. Structural design for the significant limit states

All of the following outputs can be obtained when analyzing an embankment dam using SEEP/W component of Geoslope software except.

- ☐ a. Flow net plot.
- ☐ b. Pressure head distribution along the base of the dam.
- ☐ c. Discharge through the dam body.
- ☐ d. Deformation of the dam body.

The process each manager follows during life cycle of a project is known as

- ☐ a. Project management
- ☐ b. Manager life cycle
- ☐ c. Project administration
- ☐ d. Project Management life cycle

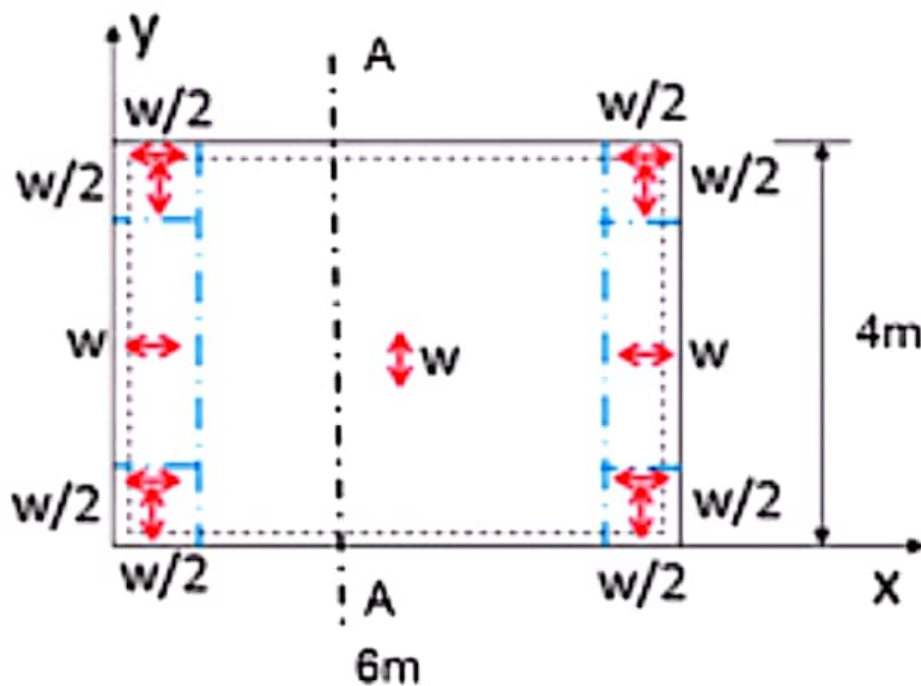
Which one of the following does NOT describe a category of constructed facilities

- ☐ a. Residential Housing Construction
- ☐ b. Institutional and Commercial Building Construction
- ☐ c. Infrastructure and heavy construction
- ☐ d. River Basins

Concrete undergoes time dependent volume change, which may cause stresses, cracking, or deflections that affect the in-service behaviour of reinforced concrete structures, these include except

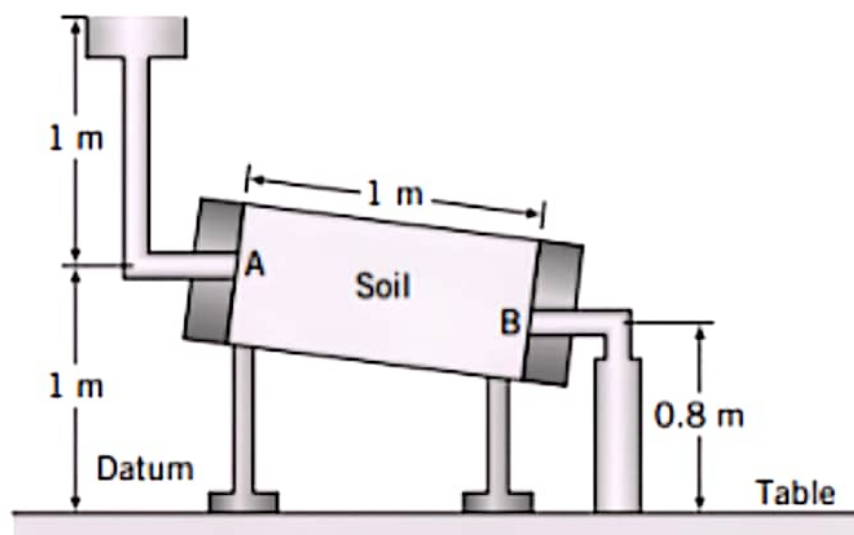
- ☐ a. Hydration
- ☐ b. Shrinkage
- ☐ c. Creep
- ☐ d. Thermal expansion or contraction

The factored load acting on the slab in figure is 10kN/m^2 . Calculate the maximum strip moment along strip A-A



- ☐ a. 20kNm/m
- ☐ b. 40kNm/m
- ☐ c. 45kNm/m
- ☐ d. 10kNm/m

A soil sample 10 cm in diameter is placed in an inclined tube 1 m long. A constant supply of water is allowed to flow into one end of the soil at A, and the outflow at B is collected by a beaker as shown in the figure. The average amount of water collected is 1 cm³ for every 10 seconds. If the void ratio is 0.6, determine the seepage velocity and the hydraulic conductivity.



- ☐ a. 0.1 cm /s, 10.8×10^{-4} cm/s
- ☐ b. 0.0034 cm/s , 10.8×10^{-4} cm/s
- ☐ c. 0.0013 cm/s, 10.8×10^{-4} cm/s
- ☐ d. 0.034 cm/s , 10.8×10^{-4} cm/s

The quality of a hardened concrete is commonly assessed through

- ☐ a. Compressive strength at the age of 3days
- ☐ b. Compressive strength at the age of 28days
- ☐ c. Compressive strength at the age of 7days
- ☐ d. Compressive strength at the age of 90days

Steel tension member is the most efficient structural element, but its efficiency may be seriously affected by

- ☐ a. Buckling
- ☐ b. Elongation
- ☐ c. Yielding
- ☐ d. End connections

The base shear computed using the equivalent static lateral force method of analysis is not dependent on

- ☐ a. The fundamental period of the building
- ☐ b. Total mass of the building
- ☐ c. Spectral acceleration of the seismic action
- ☐ d. The area of the building

A 7 m-high smooth, vertical wall retains a horizontal surface granular backfill with unit weight of 18 kN/m^3 and a friction angle of 30° . Find the magnitude and location of the resultant active thrust from the base.

- ☐ a. 147 kN/m , 2.33 m
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